



MySQL

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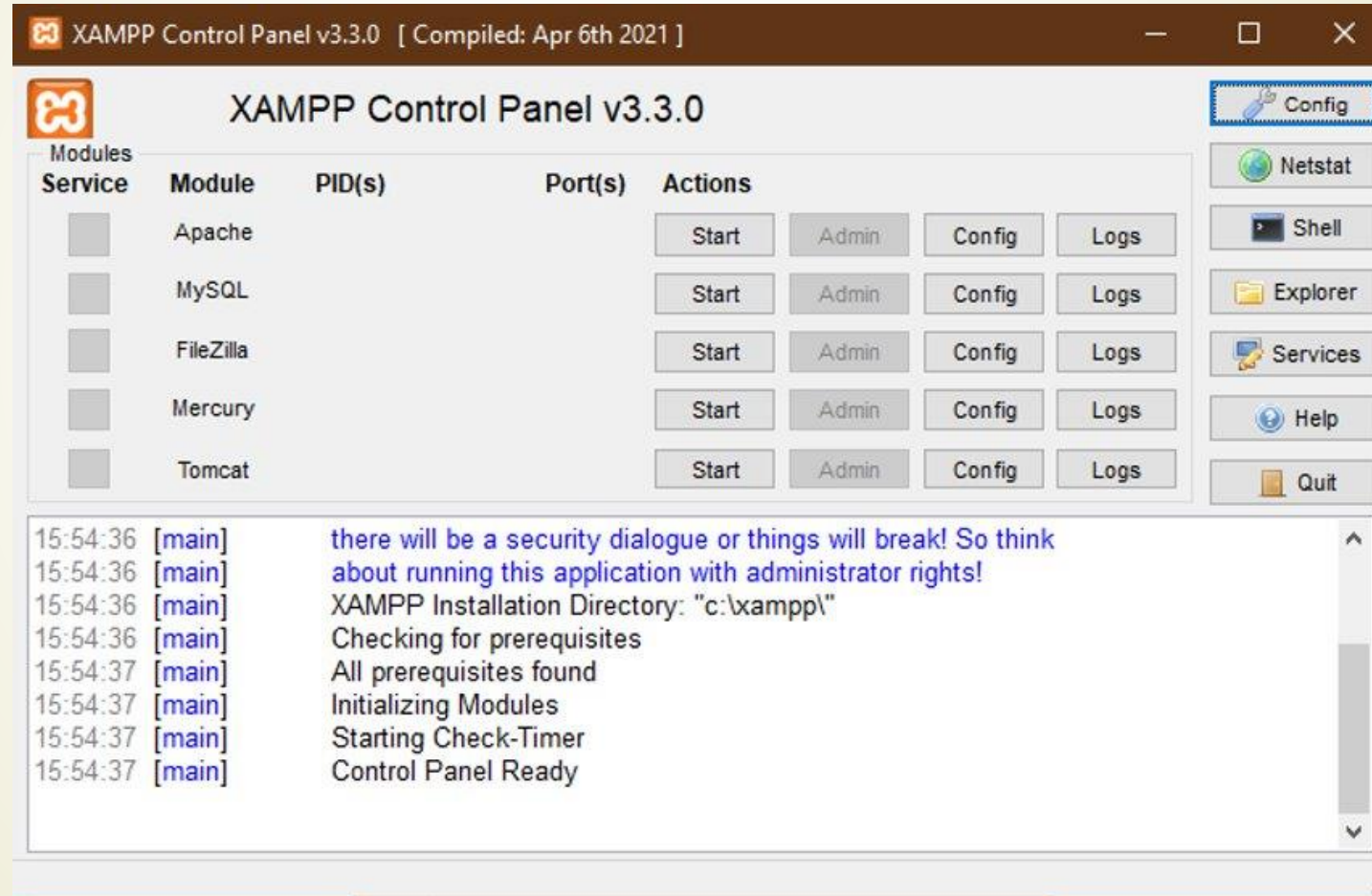


Installazione MySQL Server/Client

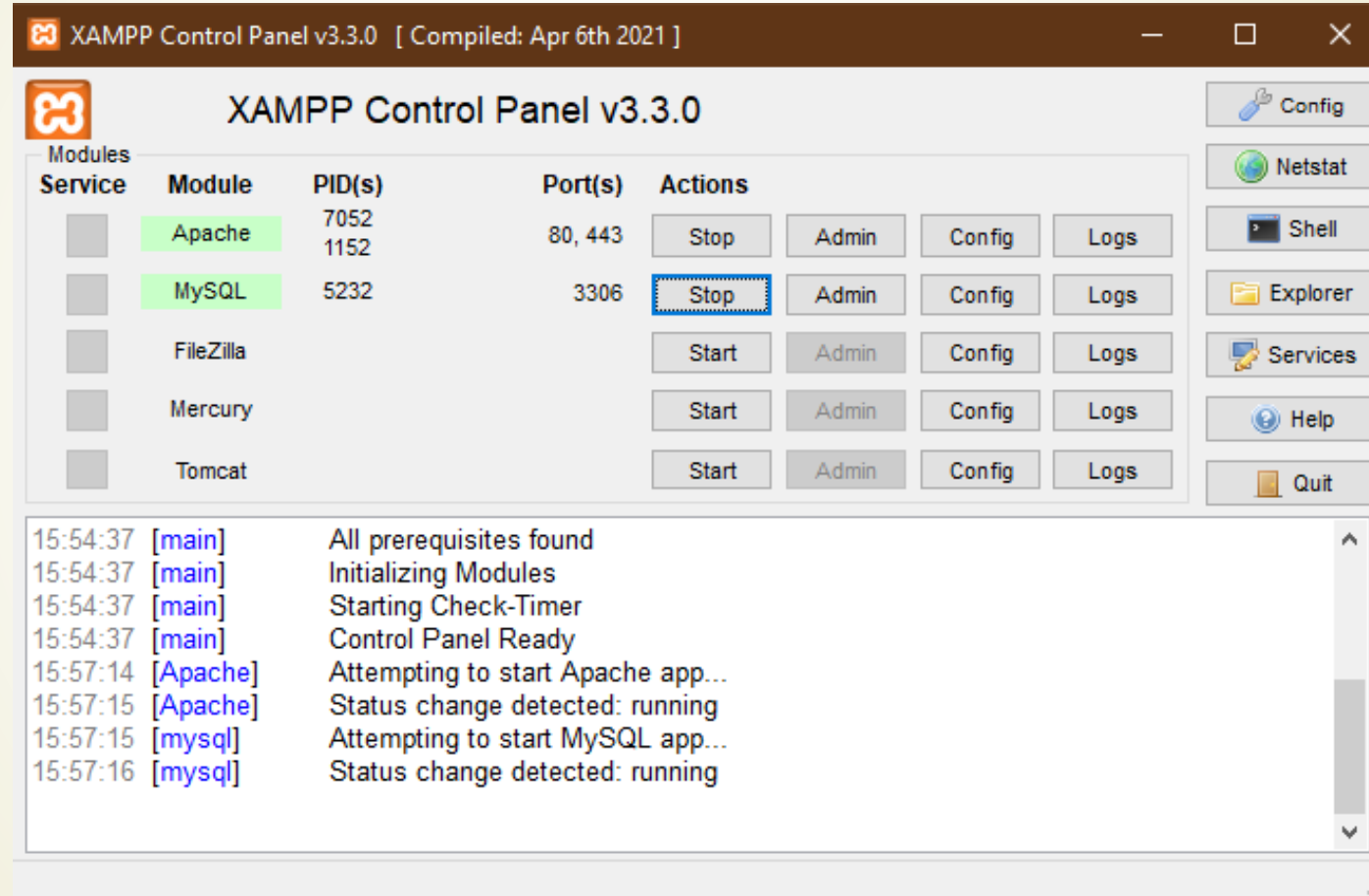
- Installare XAMPP (MySQL Server)
 - Disponibili versioni per Windows/Ubuntu/OS X

<https://www.apachefriends.org/download.html>

XAMPP Control Panel



XAMPP run services



The screenshot displays the XAMPP Control Panel v3.3.0 interface. The title bar indicates the version and compilation date: "XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]". The main window features a "Modules" section with a table listing services, their modules, PIDs, ports, and available actions. To the right of the table are buttons for "Config", "Netstat", "Shell", "Explorer", "Services", "Help", and "Quit". Below the table is a log window showing the sequence of events, including the successful start of Apache and MySQL services.

Service	Module	PID(s)	Port(s)	Actions
☐	Apache	7052 1152	80, 443	Stop Admin Config Logs
☐	MySQL	5232	3306	Stop Admin Config Logs
☐	FileZilla			Start Admin Config Logs
☐	Mercury			Start Admin Config Logs
☐	Tomcat			Start Admin Config Logs

Log entries:

- 15:54:37 [main] All prerequisites found
- 15:54:37 [main] Initializing Modules
- 15:54:37 [main] Starting Check-Timer
- 15:54:37 [main] Control Panel Ready
- 15:57:14 [Apache] Attempting to start Apache app...
- 15:57:15 [Apache] Status change detected: running
- 15:57:15 [mysql] Attempting to start MySQL app...
- 15:57:16 [mysql] Status change detected: running

Configurare ambiente MySQL

➤ Windows

- Aggiungere il percorso del binario mysql.exe ('C:\xampp\mysql\bin\') alla Variabile d'ambiente PATH

➤ Ubuntu

- Eseguire il comando

```
'export PATH $PATH:/opt/lampp/bin/'
```

ATTENZIONE in ubuntu è necessario specificare l'host a cui connettersi.

➤ MacOS

- Eseguire il comando

```
'export PATH=$PATH:/Applications/XAMPP/xamppfiles/bin'
```

Con questo comando, state aggiungendo la directory in cui si trova il file "mysql.exe" al percorso di ricerca degli eseguibili (PATH) del vostro sistema. Una volta eseguito questo comando, sarete in grado di avviare MySQL da qualsiasi posizione nel vostro sistema.

Connessione/disconnessione: riga di comando

- Per connettersi al server è necessario fornire login e password

```
mysql [-h host] [-P PortNumber] -u user -p  
Enter password: *****
```

host and *user* rappresentano:

- l'hostname dove risiede MySQL;
- lo username di un utente che possiede un account sul server;
- *-p* specifica al server la richiesta della password all'utente.

Connessione/disconnessione: riga di comando

```
C:\>mysql -u root -p
```

```
Enter password:
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 19 to server version: 4.0.18-nt
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the buffer.
```

```
mysql>
```

Help

```
C:\>mysql -help
```

Quit

```
mysql> QUIT
```



Istruzioni SQL



SELECT

Estrazione Dati

INSERT
UPDATE
DELETE

Data manipulation language (DML)

CREATE ALTER
DROP RENAME
TRUNCATE

Data definition language (DDL)

COMMIT
ROLLBACK
SAVEPOINT

controllo Transazioni

GRANT
REVOKE

Data control language (DCL)



Engine

Gli engine sono librerie sviluppate per la gestione dei dati sul disco. Il DBMS usa gli engine per le operazioni: lettura, scrittura, creazione, aggiornamento delete e indicizzazione.

- InnoDB
- MyISAM



Engine

- **InnoDB (Storage Engine di default di MySQL)**
 - Supporto alle transazioni
 - Foreign keys
 - Lock a livello di riga
 - MVCC (Multiversion Concurrency Control)
 - Clustered indexes
- **MyISAM**
 - Accesso sequenziale
 - Lock a livello di tabella



Creazione tabella InnoDB

```
CREATE TABLE opening_lines (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nome VARCHAR(20),  
    cognome VARCHAR(20),  
    ...  
  
    ) ENGINE=InnoDB;
```





Indici

- ▶ Un indice consente di recuperare più velocemente i dati di una tabella, evitandone la lettura dell'intero contenuto (full table scan).
- ▶ **CREATE INDEX** *nome_indice* **ON** *nome_tabella* (*'nome_campo'*);
- ▶ **DROP INDEX** *nome_indice* **ON** *nome_tabella*;



Indici

- ▶ PRIMARY KEY: no duplicati, no NULL
 - ▶ UNIQUE: no duplicati, Si NULL
 - ▶ COLUMN INDEX: Si duplicati e NULL
 - ▶ PREFIX INDEX: indicizza solo una porzione iniziale di caratteri.
 - ▶ FULLTEXT: ottimizza le ricerca testuale su un campo.
- 

Esempi Indici

➤ COLUMN INDEX

```
CREATE TABLE `Persone` (  
    id INT PRIMARY KEY  
    cognome VARCHAR(20) NOT NULL,  
    INDEX cognome_ind (cognome) );
```

➤ PREFIX INDEX

```
CREATE TABLE `Persone` (  
    id INT PRIMARY KEY  
    cognome VARCHAR(20) NOT NULL,  
    INDEX cognome_ind (cognome(6)) );
```



Chiavi esterne e indici

```
CREATE TABLE user(  
    iduser INT AUTO_INCREMENT,  
    name VARCHAR(50),  
    ...  
    PRIMARY KEY (iduser)) ;  
  
CREATE TABLE contact(  
    idcontact INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT,  
    tel VARCHAR(20),  
    INDEX user_ind (user_id),  
    FOREIGN KEY (user_id) REFERENCES user(iduser)) ;
```

Esempi

```
mysql> SELECT VERSION(), CURRENT_DATE;
```

```
+-----+-----+
| VERSION() | CURRENT_DATE |
+-----+-----+
| 4.0.18-nt | 2004-11-02   |
+-----+-----+
1 row in set (0.06 sec)
```

```
mysql> SELECT SIN(PI()/4), (4+1)*5;
```

```
+-----+-----+
| SIN(PI()/4) | (4+1)*5 |
+-----+-----+
| 0.707107    | 25      |
+-----+-----+
```

Esempi

```
mysql> select user()  
-> ,  
-> current_date;
```

```
+-----+-----+  
| user()          | current_date |  
+-----+-----+  
| alfredo@localhost | 2004-11-02   |  
+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql>
```

Selezione di un database

```
mysql> show databases;  
+-----+  
| Database |  
+-----+  
| alfredo  |  
| mysql    |  
| test     |  
+-----+  
3 rows in set (0.00 sec)
```

```
mysql> use alfredo;  
Database changed
```

Creazione di un database

L'amministratore crea il database;

```
mysql> CREATE DATABASE menagerie;
```

Successivamente l'utente crea le tabelle che desidera:

```
CREATE TABLE pet (name VARCHAR(20),  
                    owner VARCHAR(20),  
                    species VARCHAR(20),  
                    sex CHAR(1),  
                    birth DATE,  
                    death DATE,  
                    primary key(name,owner,birth)  
                    );
```

Describe

```
mysql> describe pet;
```

Field	Type	Null	Key	Default	Extra
name	varchar(20)	YES		NULL	
owner	varchar(20)	YES		NULL	
species	varchar(20)	YES		NULL	
sex	char(1)	YES		NULL	
birth	date	YES		NULL	
death	date	YES		NULL	

```
6 rows in set (0.02 sec)
```

Query: esempi

```
mysql> SELECT * FROM pet;
```

name	owner	species	sex	birth	death
Fluffy	Harold	cat	f	1993-02-04	NULL
Claws	Gwen	cat	m	1994-03-17	NULL
Buffy	Harold	dog	f	1989-05-13	NULL
Fang	Benny	dog	m	1990-08-27	NULL
Bowser	Diane	dog	m	1979-08-31	1995-07-29
Chirpy	Gwen	bird	f	1998-09-11	NULL
Whistler	Gwen	bird	NULL	1997-12-09	NULL
Slim	Benny	snake	m	1996-04-29	NULL
Puffball	Diane	hamster	f	1999-03-30	NULL

Query: esempi

```
mysql> SELECT * FROM pet WHERE name = 'Bowser';
```

name	owner	species	sex	birth	death
Bowser	Diane	dog	m	1989-08-31	1995-07-29

```
mysql> SELECT * FROM pet WHERE (species = 'cat' AND sex = 'm')  
-> OR (species = 'dog' AND sex = 'f');
```

name	owner	species	sex	birth	death
Claws	Gwen	cat	m	1994-03-17	NULL
Buffy	Harold	dog	f	1989-05-13	NULL

Query: esempi

```
mysql> SELECT name, species, birth FROM pet
-> WHERE species = 'dog' OR species = 'cat';
```

name	species	birth
Fluffy	cat	1993-02-04
Claws	cat	1994-03-17
Buffy	dog	1989-05-13
Fang	dog	1990-08-27
Bowser	dog	1989-08-31

Query: esempi

```
mysql> SELECT name, birth FROM pet ORDER BY birth DESC;
```

name	birth
Puffball	1999-03-30
Chirpy	1998-09-11
Whistler	1997-12-09
Slim	1996-04-29
Claws	1994-03-17
Fluffy	1993-02-04
Fang	1990-08-27
Bowser	1989-08-31
Buffy	1989-05-13

Query: esempi

```
mysql> SELECT name, species, birth FROM pet
-> ORDER BY species, birth DESC;
```

name	species	birth
Chirpy	bird	1998-09-11
Whistler	bird	1997-12-09
Claws	cat	1994-03-17
Fluffy	cat	1993-02-04
Fang	dog	1990-08-27
Bowser	dog	1989-08-31
Buffy	dog	1989-05-13
Puffball	hamster	1999-03-30
Slim	snake	1996-04-29

Query: esempi

```
mysql> SELECT name, birth, CURDATE(),  
-> (YEAR(CURDATE())-YEAR(birth))  
-> - (RIGHT(CURDATE(),5)<RIGHT(birth,5))  
-> AS age  
-> FROM pet;
```

name	birth	CURDATE()	age
Fluffy	1993-02-04	2003-08-19	10
Claws	1994-03-17	2003-08-19	9
Buffy	1989-05-13	2003-08-19	14
Fang	1990-08-27	2003-08-19	12
Bowser	1989-08-31	2003-08-19	13
Chirpy	1998-09-11	2003-08-19	4
Whistler	1997-12-09	2003-08-19	5
Slim	1996-04-29	2003-08-19	7
Puffball	1999-03-30	2003-08-19	4

Query: esempi

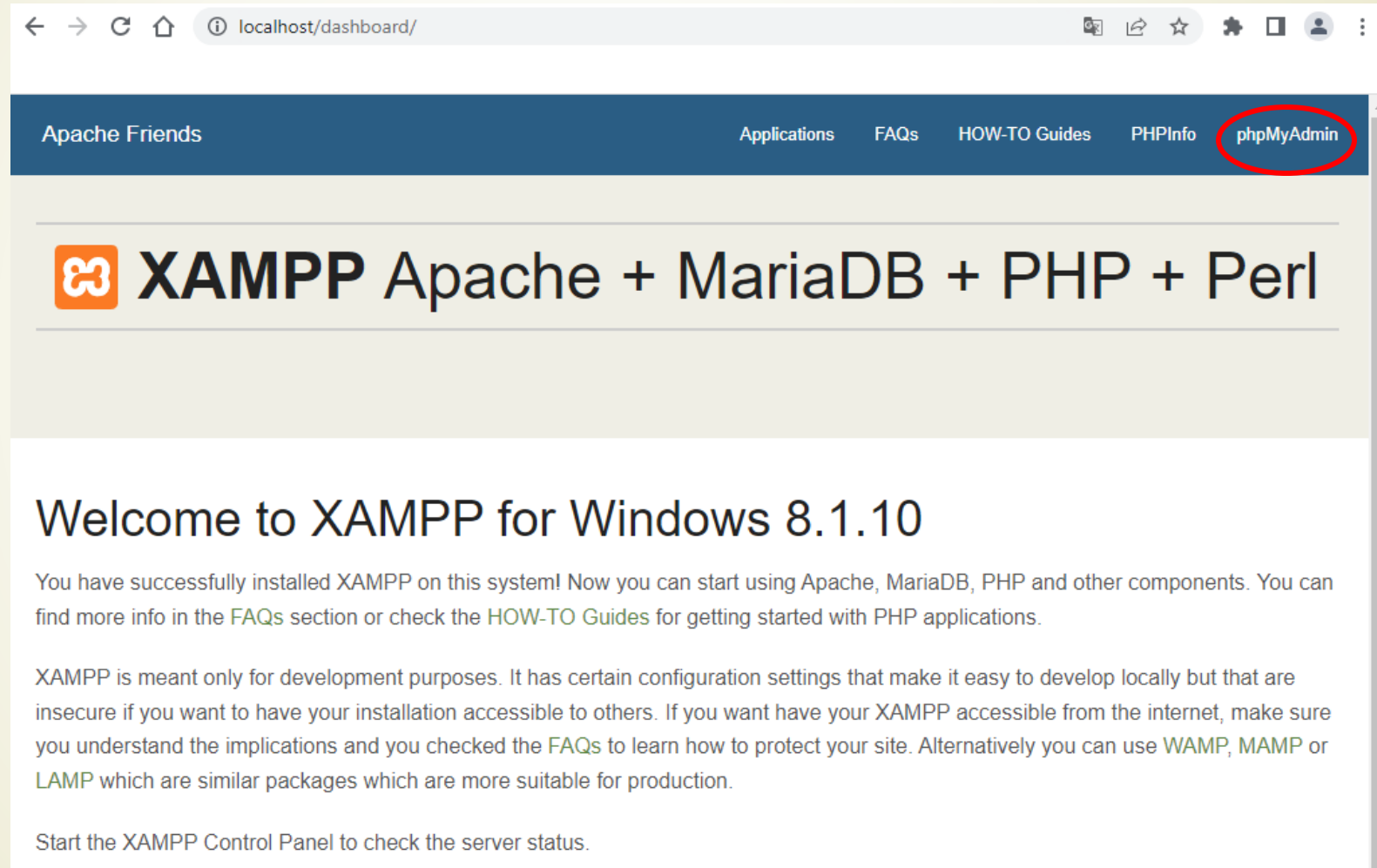
```
mysql> SELECT name, birth, death,  
-> (YEAR(death)-YEAR(birth)) - (RIGHT(death,5)<RIGHT(birth,5))  
-> AS age  
-> FROM pet  
-> WHERE death IS NOT NULL  
-> ORDER BY age;
```

name	birth	death	age
Bowser	1989-08-31	1995-07-29	5

```
mysql> SELECT name, birth FROM pet WHERE MONTH(birth) = 5;
```

name	birth
Buffy	1989-05-13

XAMPP dashboard



← → ↻ 🏠 ⓘ localhost/dashboard/ 🔍 📄 ☆ ⚙️ 🖨️ 👤 ⋮

Apache Friends Applications FAQs HOW-TO Guides PHPInfo **phpMyAdmin**

 **XAMPP** Apache + MariaDB + PHP + Perl

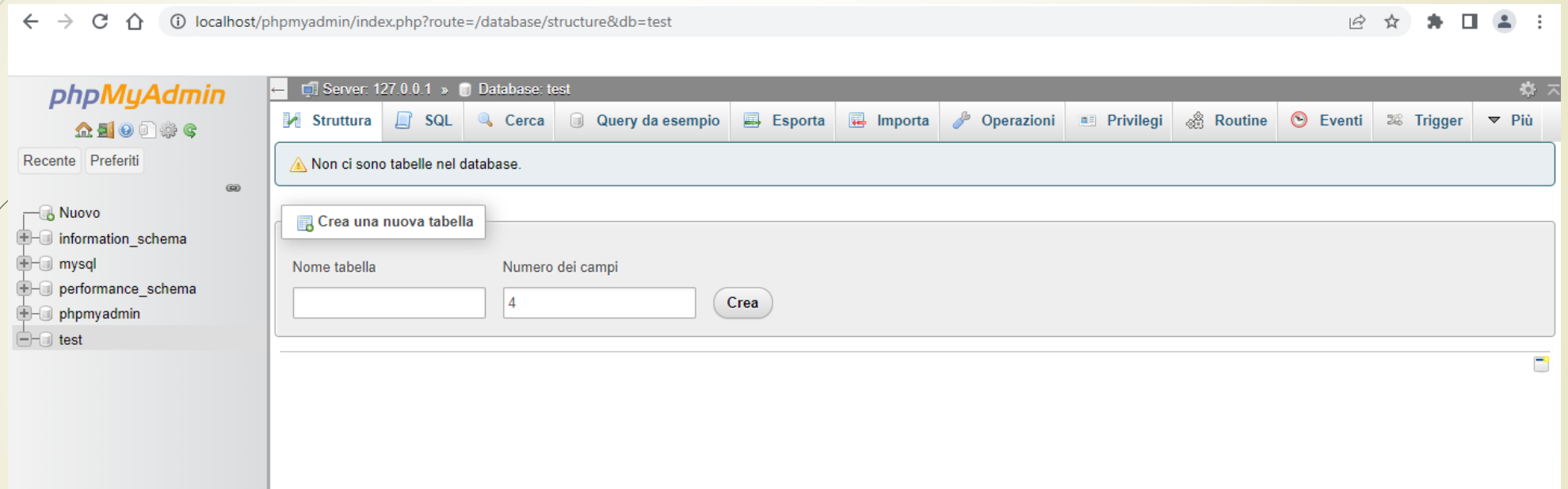
Welcome to XAMPP for Windows 8.1.10

You have successfully installed XAMPP on this system! Now you can start using Apache, MariaDB, PHP and other components. You can find more info in the [FAQs](#) section or check the [HOW-TO Guides](#) for getting started with PHP applications.

XAMPP is meant only for development purposes. It has certain configuration settings that make it easy to develop locally but that are insecure if you want to have your installation accessible to others. If you want have your XAMPP accessible from the internet, make sure you understand the implications and you checked the [FAQs](#) to learn how to protect your site. Alternatively you can use [WAMP](#), [MAMP](#) or [LAMP](#) which are similar packages which are more suitable for production.

Start the XAMPP Control Panel to check the server status.

XAMPP phpmyadmin





Installazione CLIENT SQL

- Installare HeidiSQL

- Windows

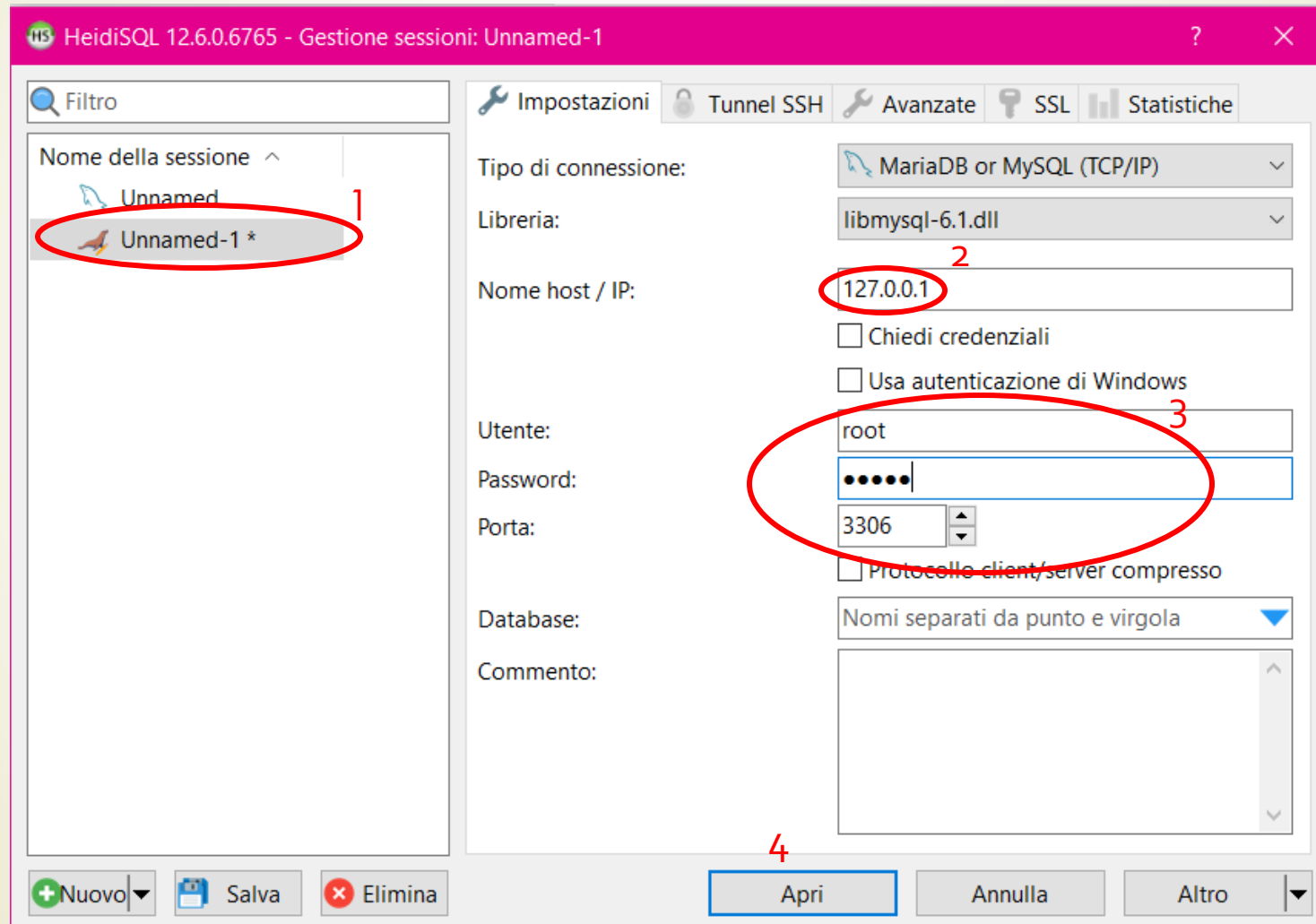
- <https://www.heidisql.com/download.php>

- Mac OS

- Sequel Pro

- <https://sequelpro.com/>

HEIDISQL CREAZIONE connessione al db



HEIDISQL

Unnamed-1\prova\pet\ - HeidiSQL 12.6.0.6765

File Modifica Cerca Query Strumenti Vai a Aiuto

Host: 127.0.0.1 Database: prova Tabella: pet

Base Opzioni Indici (1) Chiavi esterne (0) Verifica vincoli (0) Partizioni </> Codice CREATE </> Codice ALTER

Nome: pet

Commento:

Colonne: + Aggiungi ✕ Rimuovi ▲ Su ▼ Giù

#	Nome	Tipo di dati	Lunghezza/set	Senza s...	Permett...	Riem...	Predefinito	Commento	Confronto	Espressione	Virtualità
1	name	VARCHAR	20	☐	☐	☐	Nessun valore pr...		latin1_swedish_ci		
2	owner	VARCHAR	20	☐	☐	☐	Nessun valore pr...		latin1_swedish_ci		
3	species	VARCHAR	20	☐	☒	☐	NULL		latin1_swedish_ci		
4	sex	CHAR	1	☐	☒	☐	NULL		latin1_swedish_ci		
5	birth	DATE		☐	☐	☐	Nessun valore pr...				
6	death	DATE		☐	☒	☐	NULL				